SULTAN QABOOS UNIVERSITY · COLLEGE OF SCIENCE · STAT 2101

Probability and Counting Rules

Chapter 4 — the complete, exam-weighted rebuild of Dr. Ahmed Al-Hadhrami's lecture notes, mapped line-by-line to the Fall 2026 course outline.

الفصل الرابع: الاحتمالات وقواعد العدّ — ملخص شامل بالإنجليزية مع شرح عربي لكل مفهوم ومثال وقانون.

Weeks 5–6 | **Sections** 4-1 → 4-5 | **CLO** 4 (+1) | **Midterm** 3/11/2026 · 6:15–7:15 pm | **Textbook** Bluman 10e | **Target** A ≥ 90

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4-1 Sample Spaces and Probability
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DT Which rule do I use?
TR The 14 traps that cost marks
RV Review exercises, solved
A+ The A+ study protocol

00 What the outline actually tests

ما الذي يطلبه الكورس أوت لاين فعليًا من هذا الفصل

Chapter 4 is not graded on how much you can recite. The outline pins it to one learning outcome with three assessment criteria, and every quiz, midterm and final question is built from those criteria. Read this box first; it tells you where the marks live.

CHAPTER 4 IN THE COURSE SCHEDULE

WEEK	SECTIONS DELIVERED	ASSESSMENT ATTACHED

WEEK	SECTIONS DELIVERED	ASSESSMENT ATTACHED
5	4-1 Sample Spaces and Probability 4-2 The Addition Rules 4-3 Multiplication Rules & Conditional Probability *Lab 5 — RStat2101: Descriptive Measures*	Exercises Ch. 4, p. 247 `{6, 8, 16, 40}` Moodle Exercises **Lab Quiz I**
6	4-4 Counting Rules 4-5 Probability and Counting Rules	Moodle Exercises (cont.) **In-Class Quiz II**
8	—	**Midterm Exam** · 25% · 3/11/2026, Tue 6:15–7:15 pm · **Chapters 1, 2, 3, 4**
16	—	**Final Exam** · 45% · 29/12/2026, 8:00–11:00 am · all chapters

CLO4 — Apply Fundamental Probability Concepts

This is the outcome Chapter 4 serves. Its assessment criteria, verbatim from the outline:

#	CRITERION — "TO ACHIEVE THIS OBJECTIVE, THE STUDENT MUST…"	COVERED BY
A	**Calculate probabilities using counting rules and basic probability principles.**	**All of Chapter 4.** This is the one that matters here.
B	Distinguish between discrete and continuous probability distributions.	Chapter 5 — not Chapter 4.
C	Compute and interpret expected value and variance for discrete random variables.	Chapter 5 — not Chapter 4.

CLO4 is assessed by **Tutorial Questions + In-Class Quiz + Active Learning Exercise + Midterm Exam + Final Exam**. That is five separate hits. CLO1 (*Identify key statistical concepts and terms*) also runs through this chapter: definition questions — *what is a sample space, what does mutually exclusive mean* — are CLO1 marks and are the cheapest marks in the course.

THE SINGLE HIGHEST-VALUE SENTENCE IN THIS DOCUMENT

Criterion 4-A says "**counting rules and basic probability principles**" in the same breath. That is the examiner telling you in advance that §4-5 problems — where you compute `n(E)` and `n(S)` with combinations and *then* divide — are the signature Chapter 4 question. Sections 4-4 and 4-5 carry In-Class Quiz II on their own. Do not treat counting as an afterthought at the end of the chapter; it is the destination of the whole chapter.

خلاصة سريعة

الفصل الرابع كله يخدم مخرج تعلم واحد (CLO4-A): حساب الاحتمالات باستخدام قواعد العدّ والمبادئ الأساسية. يُقاس في خمسة تقييمات: التمارين، الكويز الصفي، نشاط التعلّم النشط، الميدتيرم (25%)، والنهائي (45%). الأقسام **4-4** و **4-5** (قواعد العدّ) تحمل **الكويز الصفي الثاني** بالكامل — لا تؤجلها.

Grade arithmetic: what A actually costs

The A–F scale puts **A at ≥ 90** and A− at ≥ 86. With In-Class Quizzes 15%, Lab Quiz 10%, Active Learning 5%, Midterm 25% and Final 45%, Chapter 4 content appears in at least four of those five buckets. A clean Chapter 4 is worth more than any single other chapter in the first half of the course, because it is the last chapter on the midterm *and* the foundation of Chapters 5 and 6 on the final.

لماذا هذا الفصل مهم لدرجتك؟

الـ A تبدأ من `90` . الفصل الرابع يدخل في أربعة من خمسة تقييمات على الأقل، وهو **آخر فصل في الميدتيرم**، و**الأساس الذي يُبنى عليه الفصلان الخامس والسادس** في النهائي (التوزيع ذو الحدين والتوزيع الطبيعي كلاهما احتمالات). أي ضعف هنا ينتقل معك إلى نهاية الفصل الدراسي، وأي إتقان هنا يسهّل عليك ما بعده.

4-1 Sample Spaces and Probability

فضاءات العينة والاحتمال

The whole of probability is one idea executed carefully: **name every outcome that could happen, name the ones you care about, compare the two counts.** Everything in §4-1 is machinery for doing that without losing an outcome.

1 · The five words you must define on command

These are CLO1 marks. They are asked as short definition questions and they are asked as the first part of long questions. Learn them as sentences, not as feelings.

TERM	DEFINITION (EXAM WORDING)	EXAMPLE
Probability experiment التجربة الاحتمالية	A chance process that leads to well-defined results called outcomes.	Rolling two dice.
Outcome النتيجة	The result of a *single trial* of a probability experiment.	Getting (5, 4).
Sample space `S` فضاء العينة	The set of *all possible* outcomes of a probability experiment. The number of elements in `S` is `n(S)`.	`S` = all 36 dice pairs.
Event الحدث	A set of outcomes of a probability experiment. Denoted `A, B, C…` or `E₁, E₂, E₃…`	`E` = "sum is 7".
Simple event الحدث البسيط	An event with **one** outcome.	"All three children are boys" = {BBB}.
Compound event الحدث المركّب	An event consisting of **two or more** outcomes.	"Exactly two girls" = {BGG, GBG, GGB}.

> **تمييز سريع**
>
> **Outcome** = نتيجة تجربة واحدة. **Sample space** = كل النتائج الممكنة. **Event** = مجموعة جزئية من فضاء العينة. إذا كانت تحتوي عنصرًا واحدًا فهي **simple**، وإذا احتوت عنصرين فأكثر فهي **compound**.

2 · Writing sample spaces — the three standard devices

الشرح بالعربي — كيف أكتب فضاء العينة؟

قبل حساب أي احتمال يجب أن تكتب **فضاء العينة كاملًا** دون أن تنسى أي نتيجة، لأن المقام في كل قانون هو `n(S)` .

توجد ثلاث طرق:

1. **الجدول:** مناسب لرمي حجري نرد — `6 × 6 = 36` نتيجة على شكل أزواج مرتبة.
2. **السرد المنظّم:** تكتب النتائج بترتيب ثابت (كل الأولاد أولًا، ثم بنت واحدة في كل موقع، ثم بنتان...) حتى لا تسقط نتيجة.
3. **شجرة الاحتمالات:** كل مرحلة تتفرع إلى احتمالاتها، وعدد الأطراف النهائية = عدد النتائج. الشجرة هي **الدليل** على أنك لم تنسَ شيئًا.

DEVICE 1 — THE GRID (TWO DICE)

EXAMPLE 4-1 · ROLLING DICE

Find the sample space for rolling two dice.

Each die has 6 faces, and the dice are distinguishable (one red, one white). Lay them out as a 6 × 6 grid of ordered pairs:

$$S = \{(1,1), (1,2), \ldots, (1,6), (2,1), \ldots, (6,6)\}$$

So $n(S) = 36$

Why it matters: every dice question in the chapter divides by 36. Memorise the sum distribution now — sum 7 occurs 6 times, sums 6 and 8 occur 5 times each, 5 and 9 four times, 4 and 10 three times, 3 and 11 twice, 2 and 12 once. Total 36. ✔

DEVICE 2 — SYSTEMATIC LISTING (GENDER OF CHILDREN)

EXAMPLE 4-3 · GENDER OF CHILDREN

Find the sample space for the gender of the children if a family has three children. Use B for boy and G for girl.

$$S = \{BBB, BBG, BGB, GBB, BGG, GBG, GGB, GGG\}$$

$n(S) = 8$ (= 2 × 2 × 2, by the Fundamental Counting Rule of §4-4)

Note the ordering discipline: all-boys first, then one girl in each of three positions, then two girls in each of three positions, then all girls. Listing in a fixed order is how you avoid dropping an outcome under exam pressure.

DEVICE 3 — THE TREE DIAGRAM

A **tree diagram** is a device consisting of line segments emanating from a starting point and also from the outcome point. It is used to determine all possible outcomes of a probability experiment.

EXAMPLE 4-4 · GENDER OF CHILDREN

Use a tree diagram to find the sample space for the gender of three children in a family.

Branch on the first child (B / G); from each of those, branch on the second child; from each of those, branch on the third. You get 2 × 2 × 2 = 8 terminal branches, read left to right:

FIRST CHILD	SECOND CHILD	THIRD CHILD	OUTCOME
B	B	B	BBB
B	B	G	BBG
B	G	B	BGB
B	G	G	BGG
G	B	B	GBB
G	B	G	GBG
G	G	B	GGB
G	G	G	GGG

Same eight outcomes as Example 4-3 — the tree is the *proof* that nothing was missed.

3 · Venn diagrams and set operations

A **Venn diagram** is a graphical way of representing the relationships between sets. The rectangle is S; circles inside it are events. Three standard configurations: overlapping ($A \cap B \neq \varnothing$), disjoint ($A \cap B = \varnothing$), and nested ($B \subset A$).

THE THREE SET OPERATIONS

Union $A \cup B$ — read "**A or B**" — all outcomes in either event *or both*

Intersection $A \cap B$ — read "**A and B**" — all outcomes in *both* events.

Complement $\bar{A}$ or A^c — read "**not A**" — all outcomes in S that are *no*

In probability, "or" is always *inclusive*: A or B or both. This is not the everyday English "or". It is the single most common source of lost marks in §4-2.

الشرح بالعربي — مخطط فن والعمليات على الأحداث

مخطط فن (Venn diagram): مستطيل يمثل فضاء العينة S ، وبداخله دوائر تمثل الأحداث.

- **الاتحاد** $A \cup B$ ويُقرأ **A or B**: كل العناصر الموجودة في A أو في B **أو في الاثنين معًا**.
- **التقاطع** $A \cap B$ ويُقرأ **A and B**: العناصر **المشتركة فقط** بين الحدثين.
- **المتمّمة** $\bar{A}$ أو A^c وتُقرأ **not A**: كل عناصر S التي **ليست** في A .

تنبيه: كلمة **or** في الاحتمال تعني «أحدهما أو كلاهما» دائمًا، وليست «أحدهما فقط» كما في الكلام اليومي. والمتمّمة تُحسب من داخل S فقط — في المثال أعلاه الرقم 8 يدخل في $\bar{A}$ لأنه ليس من عوامل 12.

WORKED EXAMPLE · 12-SIDED DIE

Roll a 12-sided die once. $S = \{1,\ldots,12\}$. Let A = factors of 12 and B = factors of 8. Represent S, A, B on a Venn diagram, then find (a) $A \cup B$, (b) $A \cap B$, (c) $\bar{A}$, (d) $(A \cup B)^c$.

First, build the sets. Factors of 12: 1, 2, 3, 4, 6, 12 → $A = \{1, 2, 3, 4, 6, 12\}$. Factors of 8: 1, 2, 4, 8 → $B = \{1, 2, 4, 8\}$.

Venn layout. Inside A only: 3, 6, 12. Inside both: 1, 2, 4. Inside B only: 8. Outside both, still in S: 5, 7, 9, 10, 11.

1. (a) $A \cup B = \{1, 2, 3, 4, 6, 8, 12\}$ → $n = 7$
2. (b) $A \cap B = \{1, 2, 4\}$ → $n = 3$
3. (c) $\bar{A} = \{5, 7, 8, 9, 10, 11\}$ → $n = 6$ (everything in S not in A — note 8 *is* included)
4. (d) $(A \cup B)^c = \{5, 7, 9, 10, 11\}$ → $n = 5$

Check: $n(A \cup B) + n((A \cup B)^c) = 7 + 5 = 12 = n(S)$. ✔ Always run this check — it catches a dropped element in two seconds.

TRAP — THE COMPLEMENT MUST STAY INSIDE S

$\bar{A}$ is *not* "everything else in the universe". It is everything in **this sample space** that is not in A. In the example above, students routinely forget 8 belongs in $\bar{A}$ because it "feels like it belongs to B". B is irrelevant to $\bar{A}$.

4 · The three interpretations of probability

Probability is used to quantify the likelihood, or chance, that an outcome of a random experiment will occur. The probability that event A occurs is written $P(A)$. There are three

basic interpretations:

1 · Classical

Uses sample spaces. Assumes **all outcomes are equally likely**. Computed before any data is collected — pure counting.

2 · Empirical (relative frequency)

Based on **observation**. Computed from a frequency distribution of data already collected.

3 · Subjective

A value based on an **educated guess or estimate**, employing opinions and inexact information.

الشرح بالعربي — أنواع الاحتمال الثلاثة

- **الكلاسيكي (Classical):** يُحسب من فضاء العينة **قبل** إجراء أي تجربة، بشرط أن تكون كل النتائج متساوية الفرص. مثال: احتمال ظهور 3 عند رمي نرد = 1/6 .
- **التجريبي (Empirical):** يُحسب من **بيانات جُمعت فعلًا** — جدول تكراري أو استبيان — ويساوي التكرار ÷ المجموع.
- **الشخصي (Subjective):** تقدير مبني على **رأي أو خبرة** شخص، بلا عدّ وبلا بيانات.

طريقة التمييز في الامتحان: فضاء عينة يُعدّ ← كلاسيكي. جدول أو «في استبيان شمل n شخصًا» ← تجريبي. رأي خبير ← شخصي.

CLASSICAL PROBABILITY

Equally likely events are events that have the same probability of occurring. The classical probability of any event `E` of a sample space `S` is:

CLASSICAL PROBABILITY

$$P(E) = \frac{n(E)}{n(S)}$$

`n(E)` = the number of outcomes in E. `n(S)` = the number of outcomes in S.

EXAMPLE 4-6 · GENDER OF CHILDREN

If a family has three children, find the probability that exactly two of the three children are girls.

1. **Sample space:** S = {BBB, BBG, BGB, GBB, BGG, GBG, GGB, GGG}, so n(S) = 8.
2. **Event:** E = "exactly two of the three children are girls" = {BGG, GBG, GGB}, so n(E) = 3.
3. **Divide:** P(E) = n(E)/n(S) = 3/8 = **0.375**

Important (the lecturer flags this explicitly): sample spaces and events must be *written out using set notation* as shown above. Marks are allocated to the set, not only to the final number. Writing just "3/8" loses method marks.

THE FOUR PROBABILITY RULES

PROBABILITY RULES — LET EVENT E BE IN SAMPLE SPACE S

(1) $0 \le P(E) \le 1$

(2) If $P(E) = 0$, then E **cannot happen** (impossible).

(3) If $P(E) = 1$, then E is **certain**.

(4) The sum of the probabilities of all outcomes in S equals 1.

The probability line: 0 = Impossible · ~0.25 = Unlikely · 0.5 = 50–50 · ~0.75 = Likely · 1 = Certain.

الشرح بالعربي — القانون الكلاسيكي وقواعد الاحتمال

الاحتمال الكلاسيكي = عدد النتائج التي تحقق الحدث ÷ عدد كل النتائج الممكنة: $P(E) = n(E) / n(S)$.

قواعد الاحتمال الأربع:

1. الاحتمال دائمًا بين الصفر والواحد: $0 \le P(E) \le 1$.
2. إذا كان $P(E) = 0$ فالحدث **مستحيل** (مثل ظهور 9 على نرد عادي).
3. إذا كان $P(E) = 1$ فالحدث **مؤكد** (مثل ظهور عدد أقل من 7).
4. مجموع احتمالات كل النتائج في فضاء العينة = 1 .

استخدم القاعدة الأولى لفحص كل إجابة: أي احتمال **سالب** أو **أكبر من 1** يعني أن في الحل خطأ. وتذكّر أن الدكتور يطلب كتابة فضاء العينة والحدث **بصيغة المجموعات** {...} — عليها درجات.

EXAMPLES 4-8 & 4-9 · ROLLING A DIE

(a) When a single die is rolled, find the probability of getting a 9. (b) When a single die is rolled, what is the probability of getting a number less than 7?

(a) S = {1, 2, 3, 4, 5, 6}; E: getting a 9. It is impossible to get a 9, so E = { } and n(E) = 0.

$$P(E) = 0/6 = \boxed{0}$$

(b) S = {1, 2, 3, 4, 5, 6}; E: getting a number less than 7. Every face qualifies, so E = {1, 2, 3, 4, 5, 6} and n(E) = 6.

$$P(E) = 6/6 = \boxed{1} \text{ — the event is } \textbf{certain}.$$

These two examples exist to demonstrate rules (2) and (3). An exam question that produces `P = 0` or `P = 1` is testing whether you *recognise* impossibility and certainty — say the words in your answer.

5 · Complementary events

The **complement of an event E** is the set of outcomes in the sample space that are not included in the outcomes of event E. It is denoted $\bar{E}$ (read "E bar").

RULE FOR COMPLEMENTARY EVENTS

$$P(\bar{E}) = 1 - P(E) \quad \text{or} \quad P(E) = 1 - P(\bar{E}) \quad \text{or} \quad P(E) + P(\bar{E}) = 1$$

This is the highest-leverage formula in the chapter. Whenever a question contains "**not**", "**at least one**", "**neither**", or "**none**", reach for it before you do anything else.

الشرح بالعربي — المتمّمة

متمّمة الحدث E هي كل النتائج التي **لا تحقق** E. وبما أن الحدث إما أن يقع أو لا يقع، فمجموع الاحتمالين = 1، ومنه:

$$P(\bar{E}) = 1 - P(E)$$

متى تستخدمها؟ عندما ترى في السؤال: **not** (ليس)، **neither ... nor** (لا هذا ولا ذاك)، **none** (ولا واحد)، **at least one** (واحد على الأقل). هذه القاعدة كثيرًا ما تحوّل سؤالًا طويلًا إلى عملية طرح واحدة. مثال: 23% يفضّلون الفانيلا، إذن احتمال **ليس** الفانيلا = $1 - 0.23 = 0.77$.

EXAMPLE 4-10 · FINDING COMPLEMENTS

Find the complement of each event: (a) selecting a month that has 30 days; (b) selecting a day of the week that begins with the letter S; (c) rolling two dice and getting a number whose sum is 7.

1. **(a)** Months have 28, 30 or 31 days. Complement = **selecting a month that has 28 or 31 days.**
2. **(b)** Days starting with S are Saturday and Sunday. Complement = **selecting a day that does not begin with the letter S** — that is, Monday, Tuesday, Wednesday, Thursday or Friday.
3. **(c)** The possible sums of two dice are 2 through 12. Complement = **getting a sum of 2, 3, 4, 5, 6, 8, 9, 10, 11, or 12.**

NOTE ON THE SLIDE

The lecture slide for part (b) prints "does not begin with the letter **T**", which is a typo carried from an earlier version of the example. The event is "begins with S", so its complement is "does *not* begin with S". Write S in your answer and you are correct against the textbook.

EXAMPLE 4-11 · FAVORITE ICE CREAM FLAVORS

In a study, 23% of the people surveyed said that vanilla was their favorite flavor of ice cream. If a person is selected at random, find the probability that the person's favorite flavor is *not* vanilla.

Let E be "favorite flavor is vanilla", with P(E) = 0.23.

$$P(\text{not vanilla}) = 1 - P(E) = 1 - 0.23 = \boxed{0.77} \quad (\text{or } 77\%)$$

6 · Empirical probability

Given a frequency distribution, the probability of an event being in a given class is the relative frequency of that class:

EMPIRICAL (RELATIVE FREQUENCY) PROBABILITY

$$P(E) = \frac{\text{frequency for the class}}{\text{total frequencies in the distribution}} = \frac{f}{n}$$

`f` = frequency of the class (event); `n` = the sample size. Based on **observation**, not on assuming equally likely outcomes.

الشرح بالعربي — الاحتمال التجريبي

يُحسب من البيانات المجموعة: `P(E) = f / n` ، حيث `f` تكرار الفئة و `n` حجم العينة الكلي. مثال: سألت 50 شخصًا وأجاب 3 منهم «قطار أو حافلة»، فالاحتمال `3/50 = 0.06` .

نصيحة: اجمع عمود التكرارات وتأكد أنه يساوي `n` ، واجمع عمود الاحتمالات وتأكد أنه يساوي `1` . وإذا طُلب احتمال فئتين معًا (A أو B) فاجمع تكراريهما ثم اقسم.

EXAMPLE 4-12 · TRAVEL SURVEY

A researcher asked 50 people who plan to travel over the Thanksgiving holiday how they will get to their destination: Drive 41, Fly 6, Train or bus 3. Find the probability that a person will travel by train or bus.

METHOD	FREQUENCY F	EMPIRICAL PROBABILITY F/N
Drive	41	41/50 = 0.82
Fly	6	6/50 = 0.12
Train or bus	3	3/50 = 0.06
Total	**50**	**50/50 = 1.00**

`P(train or bus) = f/n = 3/50 =` **0.06**

The total column summing to exactly 1.00 is probability rule (4) in action — use it as a check on every frequency table you build.

EXAMPLE 4-13 · DISTRIBUTION OF BLOOD TYPES

In a sample of 50 people, 21 had type O blood, 22 had type A, 5 had type B, and 2 had type AB. Set up a frequency distribution and find: (a) P(type O); (b) P(type A or type B); (c) P(neither type A nor type O); (d) P(not type AB).

TYPE	FREQUENCY
A	22
B	5
AB	2
O	21
Total	**50**

1. **(a)** P(O) = f/n = 21/50 = **0.42**
2. **(b)** P(A or B) = 22/50 + 5/50 = 27/50 = **0.54** (add the frequencies of the two classes — the types are mutually exclusive)
3. **(c)** "Neither A nor O" means the person has type B or type AB: P = 5/50 + 2/50 = 7/50 = **0.14**
4. **(d)** P(not AB) = 1 − P(AB) = 1 − 2/50 = 48/50 = **0.96**

Read the structure: (b) is the addition rule, (c) is a translated complement ("neither…nor" = complement of a union), (d) is the complement rule. One table, three different tools. This is exactly how a long exam question is built.

EXAMPLE 4-14 · SLEEP HOURS

A survey found the following distribution for hours of sleep per night: Less than 6 h → 12; 6 h → 26; 7 h → 30; 8 h → 28; More than 8 h → 4. Total 100. Find, for a person selected at random: (a) sleeps 8 hours; (b) sleeps fewer than 7 hours; (c) sleeps **at most** 8 hours; (d) sleeps **at least** 7 hours.

1. **(a) Exactly 8 hours** — one class only: P = 28/100 = **0.28**
2. **(b) Fewer than 7 hours** — strictly below 7, so "less than 6" and "6": P = (12 + 26)/100 = 38/100 = **0.38**
3. **(c) At most 8 hours** — 8 and everything below it: P = (12 + 26 + 30 + 28)/100 = 96/100 = **0.96**
   Faster: 1 − P(more than 8) = 1 − 4/100 = 0.96 ✔
4. **(d) At least 7 hours** — 7 and everything above it: P = (30 + 28 + 4)/100 = 62/100 = **0.62**

7 · The five phrases — translate before you calculate

The lecturer gives this as a standalone "Important Note" slide, which is a strong signal that it is examined. Consider the values **3, 4, 5, 6, 7**:

PHRASE	VALUES USED	SYMBOL
Exactly 5	5	$X = 5$
Less (fewer) than 5	3, 4	$X < 5$ — 5 is *excluded*
At most 5 (5 or less)	3, 4, 5	$X \leq 5$ — 5 is *included*
More (greater) than 5	6, 7	$X > 5$ — 5 is *excluded*
At least 5	5, 6, 7	$X \geq 5$ — 5 is *included*

الترجمة الحرفية — احفظها

- **Exactly 5** = بالضبط 5 فقط.
- **Less / fewer than 5** = أقل من 5، **بدون** الـ 5.
- **At most 5** = 5 فأقل، **مع** الـ 5.
- **More / greater than 5** = أكثر من 5، **بدون** الـ 5.
- **At least 5** = 5 فأكثر، **مع** الـ 5.

القاعدة الذهبية: كلمة (at most / at least) **at** تعني أن الرقم نفسه **داخل** الحساب. وكلمة (less than / more than) تعني أن الرقم نفسه **خارج** الحساب.

8 · Subjective probability

Subjective probability uses a probability value based on an educated guess or estimate, employing opinions and inexact information. No sample space is counted and no data is tabulated — a qualified person simply assigns a number.

- A sportswriter may say there is a **70% probability** that the Oman football team will win the Gulf Cup next year.
- A physician might say that, on the basis of her diagnosis, there is a **30% chance** the patient will need an operation.
- A seismologist might say there is an **80% probability** that an earthquake will occur in a certain area.

الشرح بالعربي — الاحتمال الشخصي

رقم يعطيه شخص بناءً على **خبرته أو رأيه**، مثل قول محلل رياضي إن احتمال فوز منتخب عُمان بكأس الخليج 70% ، أو قول طبيب إن احتمال حاجة المريض لعملية 30% . لا يوجد فضاء عينة يُعدّ ولا جدول بيانات — مجرد تقدير مدروس. غالبًا يأتي في الامتحان كسؤال: «ما نوع هذا الاحتمال؟»

HOW THIS IS EXAMINED

You will be shown a statement and asked *which type of probability is this?* The tell: counting a sample space → **classical**. A frequency table or "in a survey of n people" → **empirical**. An expert's opinion with no data behind it → **subjective**. One mark, two seconds, never miss it.

4-2 The Addition Rules for Probability

قواعد الجمع للاحتمال

§4-2 answers exactly one question: **what is P(A *or* B)?** There are two formulas, and choosing between them depends on a single yes/no test.

1 · Mutually exclusive events — the test

Two events are said to be **mutually exclusive events** (or **disjoint**) if they **cannot occur at the same time** — that is, they have no elements in common.

THE TEST

$$A \text{ and } B \text{ are mutually exclusive} \iff A \cap B = \varnothing$$

$\varnothing$ is the null (empty) set. On a Venn diagram, mutually exclusive events are drawn as two circles that **do not overlap**.

الشرح بالعربي — الأحداث المتنافية

الحدثان المتنافيان (**mutually exclusive** أو **disjoint**) هما حدثان **لا يمكن أن يقعا في نفس الوقت**، أي لا يوجد بينهما أي عنصر مشترك: $A \cap B = \varnothing$. على مخطط فن تكون الدائرتان منفصلتين تمامًا.

مثال: عند رمي نرد، «عدد زوجي» $\{2, 4, 6\}$ و«عدد فردي» $\{1, 3, 5\}$ متنافيان. أما «عدد زوجي» و«عامل من عوامل 4» $\{1, 2, 4\}$ فغير متنافيين لأن 2 و 4 مشتركان.

طريقة الإجابة الكاملة: اكتب التقاطع صراحةً ثم احكم — لا تكتفِ بـ «نعم» أو «لا».

WORKED EXAMPLE · ROLLING ONE DIE

S = {1, 2, 3, 4, 5, 6}. A = getting an even number = {2, 4, 6}; B = getting an odd number = {1, 3, 5}; C = getting a factor of 4 = {1, 2, 4}. (a) Are A and B mutually exclusive? (b) Are A and C mutually exclusive?

1. **(a)** $A \cap B = \{2,4,6\} \cap \{1,3,5\} = \varnothing$. **Yes — mutually exclusive.** A number cannot be even and odd at once.
2. **(b)** $A \cap C = \{2,4,6\} \cap \{1,2,4\} = \{2, 4\} \neq \varnothing$. **No — not mutually exclusive.** Rolling a 2 or a 4 satisfies both events simultaneously.

Method mark: always write the intersection out before you answer. "Yes/No" alone is worth less than "$A \cap C = \{2, 4\} \neq \varnothing$, therefore not mutually exclusive."

EXAMPLE 4-15 · DETERMINING MUTUALLY EXCLUSIVE EVENTS

Determine whether the two events are mutually exclusive, and explain your answer.
(a) Randomly selecting a female student / randomly selecting a student who is a second year.
(b) Randomly selecting a person with type A blood / randomly selecting a person with type O blood.
(c) Rolling a die and getting an odd number / rolling a die and getting a number less than 3.
(d) Randomly selecting a person who is under 21 years of age / randomly selecting a person who is over 30 years of age.

1. **(a)** **Not mutually exclusive.** A student can be female *and* a second year at the same time.
2. **(b)** **Mutually exclusive.** A person has exactly one blood type; being type A excludes being type O.
3. **(c)** **Not mutually exclusive.** Odd = {1, 3, 5}; less than 3 = {1, 2}. They share the outcome 1.
4. **(d)** **Mutually exclusive.** No one is simultaneously under 21 and over 30.

2 · The two addition rules

ADDITION RULE #1 — A AND B MUTUALLY EXCLUSIVE

$$P(A \cup B) = P(A) + P(B)$$

ADDITION RULE #2 — A AND B NOT MUTUALLY EXCLUSIVE

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

The overlap gets counted once in P(A) and once again in P(B), so it must be subtracted once to be counted correctly. On the Venn diagram this is the shaded lens in the middle.

الشرح بالعربي — قاعدتا الجمع

قاعدة الجمع تجيب عن سؤال واحد: **ما احتمال A أو B؟**

- **إذا كان الحدثان متنافيين:** نجمع مباشرة `P(A) + P(B)` .
- **إذا كانا غير متنافيين:** نجمع ثم **نطرح التقاطع مرة واحدة**: `P(A) + P(B) − P(A ∩ B)` ، لأن المنطقة المشتركة حُسبت مرتين — مرة داخل A ومرة داخل B.

مثال: 8 ممرضين و 5 أطباء؛ منهم ممرض واحد ذكر وطبيبان ذكور. احتمال (ممرض أو ذكر) = `8/13 + 3/13 − 1/13 = 10/13` ، لأن الممرض الذكر دخل في العدّ مرتين.

USE RULE #2 ALWAYS — IT IS SAFE

Rule #1 is a special case of Rule #2, because when A and B are mutually exclusive, `P(A ∩ B) = 0` and the subtraction disappears. If you are ever unsure under exam pressure, write Rule #2 and show that the intersection is zero. You cannot be wrong, and you gain a method mark for justifying it.

EXAMPLE 4-17 · ENDANGERED SPECIES

In the United States there are 59 species of mammals that are endangered, 75 species of birds, and 68 species of fish. If one animal is selected at random, find the probability that it is either a mammal or a fish.

1. **Total:** n(S) = 59 + 75 + 68 = 202.
2. **Mutually exclusive?** An animal cannot be both a mammal and a fish → **yes**, so use Rule #1.
3. $P(\text{mammal} \cup \text{fish}) = \frac{59}{202} + \frac{68}{202} = \frac{127}{202} \approx \boxed{0.629}$

The 75 birds appear *only* in the denominator. A classic distractor: students who ignore the birds get 127/127 = 1.

EXAMPLE 4-18 · RESEARCH AND DEVELOPMENT EMPLOYEES

Corporate research and development centers for three local companies have these numbers of employees: U.S. Steel 110, Alcoa 750, Bayer Material Science 250. If a research employee is selected at random, find the probability that the employee is employed by U.S. Steel or Alcoa.

1. **Total:** n(S) = 110 + 750 + 250 = 1110.
2. **Mutually exclusive?** An employee works for one company → **yes**, Rule #1.
3. $P(\text{Steel} \cup \text{Alcoa}) = \frac{110}{1110} + \frac{750}{1110} = \frac{860}{1110} = \frac{86}{111} \approx$ **0.775**

EXAMPLE 4-21 · SELECTING A MEDICAL STAFF PERSON

In a hospital unit there are 8 nurses and 5 physicians; 7 nurses and 3 physicians are females. If a staff person is selected, find the probability that the subject is a nurse **or** a male.

Build the two-way table first — always.

STAFF	FEMALE	MALE	TOTAL
Nurses	7	1	8
Physicians	3	2	5
Total	**10**	**3**	**13**

1. Male nurses = 8 − 7 = 1. Male physicians = 5 − 3 = 2. Total males = 3. n(S) = 13.
2. **Mutually exclusive?** No — a male nurse is both. So use Rule #2.
3. $P(N \cup M) = \frac{8}{13} + \frac{3}{13} - \frac{1}{13} = \frac{10}{13} \approx$ **0.769**

Sanity check by direct count: nurses-or-males = 7 female nurses + 1 male nurse + 2 male physicians = 10 people out of 13. ✔ Identical. Run this check whenever the numbers are small.

EXAMPLE 4-22 · DRIVING WHILE DEPRESSED

On New Year's night, the probability of a person driving while depressed is 0.32, the probability of a person having a driving accident is 0.09, and the probability of a person having a driving accident while depressed is 0.06. What is the probability of a person driving while depressed **or** having a driving accident?

1. **Extract:** P(D) = 0.32, P(A) = 0.09, P(D ∩ A) = 0.06.
2. **Mutually exclusive?** No — P(D ∩ A) = 0.06 ≠ 0, so they overlap. Rule #2.
3. $P(D \cup A) = 0.32 + 0.09 - 0.06 =$ **0.35**

TRAP — "ACCIDENT *WHILE* DEPRESSED" IS AN INTERSECTION, NOT A CONDITIONAL

The phrase "having a driving accident **while** depressed" means both things happen to the same person: $P(D \cap A) = 0.06$. It does *not* mean $P(A \mid D)$. If it had been conditional, the problem would say "*given that* the person is depressed" or "*of those who are depressed*". Misreading this turns 0.35 into a wrong answer, so slow down on the word that joins the two events.

3 · Extended addition rules (three or more events)

THREE MUTUALLY EXCLUSIVE EVENTS A, B, C

$$P(A \cup B \cup C) = P(A) + P(B) + P(C)$$

THREE EVENTS THAT ARE NOT MUTUALLY EXCLUSIVE

$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C) - P(B \cap C) + P$$

Pattern: add the singles, subtract the pairs, add back the triple. The lecture notes point you to Exercises 19–21 for the mutually exclusive version and Exercise 24 for the general version.

متى أستخدم أي قاعدة؟

اسأل نفسك سؤالًا واحدًا: **هل يمكن للحدثين أن يقعا معًا؟**

- **لا** ← متنافيان (mutually exclusive) ← **P(A) + P(B)**
- **نعم** ← غير متنافيين ← **P(A) + P(B) − P(A∩B)**

وإذا ترددت، استخدم القاعدة الثانية دائمًا — فهي تعطي النتيجة الصحيحة في الحالتين لأن التقاطع يساوي صفرًا عند التنافي.

4-3 The Multiplication Rules and Conditional Probability

قواعد الضرب والاحتمال الشرطي

§4-2 handled "**or**". §4-3 handles "**and**" — and then inverts it to handle "**given that**". This is the densest section in the chapter and the one that separates a B from an A.

1 · Independent vs dependent events

Independent events

Two events `A` and `B` are **independent** if the fact that A occurs **does not affect** the probability of B occurring.

Dependent events

Two events `A` and `B` are **dependent** if the fact that A occurs **does affect** the probability of B occurring.

IMPORTANT NOTE — INDEPENDENT ≠ MUTUALLY EXCLUSIVE

These two ideas are constantly confused and the lecturer calls it out explicitly. When tossing a **single** coin, the result can be either heads or tails but **cannot be both** — those are *mutually exclusive* events. When tossing **two** coins, the result of one flip **does not affect** the result of the other — those are *independent* events.

The deeper point: mutually exclusive events (with non-zero probabilities) are in fact *dependent*. If A happens, you know for certain B did not — knowing about A changed everything about B. Mutually exclusive is about *overlap*; independent is about *influence*.

الشرح بالعربي — أهم فكرة في القسم: الاستقلال والتبعية

- **المستقلان (independent):** وقوع أحدهما **لا يغيّر** احتمال الآخر. مثال: رمي عملتين — نتيجة الأولى لا تؤثر في الثانية.
- **التابعان (dependent):** وقوع أحدهما **يغيّر** احتمال الآخر. مثال: سحب كرتين بدون إرجاع — السحبة الأولى تغيّر ما تبقى في الصندوق.

لا تخلط بين المفهومين: «متنافيان» سؤاله «هل يوجد تداخل؟»، و«مستقلان» سؤاله «هل يؤثر أحدهما في الآخر؟». بل إن الحدثين المتنافيين (باحتمالات غير صفرية) يكونان **تابعين**، لأن وقوع A يجعل وقوع B مستحيلًا — أي أنه أثّر فيه بالكامل.

SPOTTING DEPENDENCE IN A WORD PROBLEM

- **"...and then it is replaced"** → independent. The pool resets, so probabilities repeat.
- **"...without replacement"** → dependent. Both the numerator and the denominator shrink on the second draw.
- **Separate devices** (a coin *and* a die, three different people chosen from a large population) → independent.
- **"Of these clients...", "of those who..."** → dependent; you are being given a conditional probability.

كيف أعرف من صياغة السؤال؟

- **then it is replaced** / **with replacement** (مع الإرجاع) ← **مستقلان**، الاحتمال يتكرر كما هو.
- **without replacement** (بدون إرجاع) ← **تابعان**، البسط والمقام يتغيران في السحبة الثانية.
- أدوات منفصلة (عملة ونرد، أشخاص من مجتمع كبير) ← **مستقلان**.
- **of these** / **of those who** (من بين هؤلاء) ← هذا **احتمال شرطي**، والحدثان تابعان.

2 · The multiplication rules

MULTIPLICATION RULE #1 — INDEPENDENT EVENTS

$$P(A \cap B) = P(A) \cdot P(B)$$

MULTIPLICATION RULE #2 — DEPENDENT EVENTS

$$P(A \cap B) = P(A \mid B) \cdot P(B)$$

$$P(A \cap B) = P(B \mid A) \cdot P(A)$$

Both forms are correct — use whichever conditional probability the question hands you. In practice the question gives one unconditional probability and one conditional one; multiply them.

الشرح بالعربي — قاعدتا الضرب

قاعدة الضرب تجيب عن سؤال: **ما احتمال A و B معًا؟**

- **إذا كانا مستقلين:** $P(A \cap B) = P(A) \times P(B)$. مثال: صورة في العملة و 4 في النرد = $1/2 \times 1/6 = 1/12$.
- **إذا كانا تابعين:** $P(A \cap B) = P(A) \times P(B \mid A)$ ، أي احتمال الأول × احتمال الثاني **بشرط** وقوع الأول. مثال: 53% لديهم تأمين منزل، و**من بين هؤلاء** 27% لديهم تأمين سيارة ← $0.53 \times 0.27 = 0.1431$.

القاعدة الذهبية: and تعني **ضرب**، و **or** تعني **جمع**.

EXAMPLE 4-23 · TOSSING A COIN

A coin is flipped and a die is rolled. Find the probability of getting a head on the coin *and* a 4 on the die.

1. The coin and the die do not influence each other → **independent** → Rule #1.
2. P(head) = 1/2 and P(4) = 1/6.
3. $$P(\text{head} \cap 4) = \frac{1}{2} \cdot \frac{1}{6} = \frac{1}{12} \approx \boxed{0.0833}$$

Verify by sample space: n(S) = 2 × 6 = 12 outcomes, and only (H, 4) qualifies → 1/12. ✔

EXAMPLE 4-25 · SELECTING A COLORED BALL

An urn contains 3 red balls, 2 blue balls, and 5 white balls. A ball is selected and its color noted. **Then it is replaced.** A second ball is selected and its color noted. Find the probability of: (a) selecting 2 blue balls; (b) selecting 1 blue ball and then 1 white ball; (c) selecting 1 red ball and then 1 blue ball.

Total balls = 3 + 2 + 5 = 10. The ball is *replaced*, so every draw sees the same 10 balls → **independent** → Rule #1.

1. **(a)** $P(B \cap B) = \frac{2}{10} \cdot \frac{2}{10} = \frac{4}{100} = \frac{1}{25} =$ **0.04**
2. **(b)** $P(B \cap W) = \frac{2}{10} \cdot \frac{5}{10} = \frac{10}{100} =$ **0.10**
3. **(c)** $P(R \cap B) = \frac{3}{10} \cdot \frac{2}{10} = \frac{6}{100} =$ **0.06**

THE VARIANT YOU MUST BE READY FOR

If the same question said "**without replacement**", every second denominator becomes 9 and the matching numerator drops by one where the colors agree:
(a) (2/10)(1/9) = 2/90 ≈ 0.0222 · (b) (2/10)(5/9) = 10/90 ≈ 0.1111 · (c) (3/10)(2/9) = 6/90 ≈ 0.0667.
Examiners pair these two versions constantly. Circle the words "replaced" or "without replacement" the moment you see them.

EXAMPLE 4-26 · SURVEY ON STRESS

It was found that 3 out of every 4 American workers say they suffer great stress at least once a week. If three people are selected at random, find the probability that all three will say that they suffer great stress at least once a week.

1. P(stress) = 3/4 = 0.75 for each person; three people drawn from a large population → **independent**.
2. $P(\text{all three}) = \frac{3}{4} \cdot \frac{3}{4} \cdot \frac{3}{4} = \frac{27}{64} \approx$ **0.4219**

EXAMPLE 4-27 · MALE COLOR BLINDNESS

Approximately 9% of men have a type of color blindness that prevents them from distinguishing between red and green. If 3 men are selected at random, find the probability that all of them will have this type of red-green color blindness.

1. P(color blind) = 0.09; three independent selections.
2. $P(\text{all three}) = (0.09)^3 = 0.09 \times 0.09 \times 0.09 =$ **0.000729**

Interpretation marks: a result this small should be read aloud in your answer — "about 7 in 10,000 groups of three men". Interpretation is explicitly in CLO8 and costs you nothing to add.

EXAMPLE 4-29 · HOMEOWNER'S AND AUTOMOBILE INSURANCE

World Wide Insurance Company found that 53% of the residents of a city had homeowner's insurance (`H`) with the company. **Of these clients**, 27% also had automobile insurance (`A`). If a resident is selected at random, find the probability that the resident has both homeowner's and automobile insurance with World Wide.

1. **Extract carefully.** P(H) = 0.53. The phrase "*of these clients*" makes the 27% conditional: P(A | H) = 0.27.
2. Dependent → Rule #2.
3. $P(H \cap A) = P(H) \cdot P(A \mid H) = 0.53 \times 0.27 =$ **0.1431**

TRAP — "OF THESE" IS THE CONDITIONAL SIGNAL

If you read the 27% as unconditional P(A) and multiply 0.53 × 0.27 anyway, you get the right number for the wrong reason and will lose the method mark. Worse, on a question where the two routes differ you will get the wrong number. Train yourself to annotate the given values as `P(H) = 0.53` and `P(A|H) = 0.27` *before* calculating.

EXAMPLE 4-28 · UNEMPLOYED WORKERS

For a specific year, 5.2% of U.S. workers were unemployed. During that time, 33% of those who were unemployed received unemployment benefits. If a person is selected at random, find the probability that she or he received unemployment benefits if the person is unemployed.

Extract: P(U) = 0.052, and "33% *of those who were unemployed*" gives P(B | U) = 0.33.

As worded on the slide ("...*if the person is unemployed*"), the question asks for the conditional probability, which is handed to you directly:

$$P(B \mid U) = \boxed{0.33}$$

The textbook version of this example asks for the probability that a randomly selected person "was unemployed **and** received benefits", which is the multiplication rule:

$$P(U \cap B) = P(U) \cdot P(B \mid U) = 0.052 \times 0.33 = \boxed{0.01716 \approx 0.0172}$$

BE READY FOR BOTH

This pair is the clearest illustration in the chapter of how one word changes the whole calculation. "**And**" → multiply the two values (0.0172). "**Given / if / of those who**" → the conditional is already there (0.33). In an exam, write which one you are computing in words before you compute it.

3 · Tree diagrams with branch probabilities

When an experiment happens in stages and the second stage depends on the first, a tree diagram is not optional decoration — it is the method. Multiply *along* branches, add *across* the branches that satisfy the event.

الشرح بالعربي — شجرة الاحتمالات

عندما تتم التجربة على **مراحل** (مثلًا: رمي عملة لاختيار صندوق، ثم سحب كرة منه)، ارسم شجرة:

- **على طول الفرع:** اضرب الاحتمالات — هذا احتمال مسار واحد.
- **بين الفروع:** اجمع احتمالات كل المسارات التي تحقق الحدث المطلوب.
- **للتأكد:** مجموع احتمالات كل المسارات النهائية يجب أن يساوي 1 .

EXAMPLE 4-31 · SELECTING COLORED BALLS

Box 1 contains 2 red balls and 1 blue ball. Box 2 contains 3 blue balls and 1 red ball. A coin is tossed: heads → box 1 is selected and a ball is drawn; tails → box 2 is selected and a ball is drawn. Find the probability of selecting a red ball.

PATH	P(BOX)	P(BALL | BOX)	PRODUCT
Box 1 → Red	1/2	2/3	2/6 = 1/3
Box 1 → Blue	1/2	1/3	1/6
Box 2 → Red	1/2	1/4	1/8
Box 2 → Blue	1/2	3/4	3/8

Red can be reached by two different paths, so add them:

$$P(\text{Red}) = \frac{1}{2}\cdot\frac{2}{3} + \frac{1}{2}\cdot\frac{1}{4} = \frac{2}{6} + \frac{1}{8} = \frac{8}{24} + \frac{3}{24} = \frac{11}{24} \approx \mathbf{0.4583}$$

Check: all four products sum to 1/3 + 1/6 + 1/8 + 3/8 = 8/24 + 4/24 + 3/24 + 9/24 = 24/24 = 1. ✔ A tree whose branches don't sum to 1 has an error in it.

4 · Conditional probability

The **conditional probability** of an event `B` in relationship to an event `A` is the probability that event B occurs *after* event A has already occurred.

CONDITIONAL PROBABILITY

$$P(B \mid A) = \frac{P(A \cap B)}{P(A)} \qquad P(A) \neq 0$$

Read it as: *the condition goes in the denominator.* Conditioning on A means you throw away everything outside A and treat A as the new sample space.

CONDITIONAL PROBABILITY AND INDEPENDENCE

If A and B are independent: $P(B \mid A) = \dfrac{P(A \cap B)}{P(A)} = P(B)$

Knowing A happened tells you nothing new about B — which is exactly what "independent" means. This doubles as the **test for independence**.

الشرح بالعربي — الاحتمال الشرطي

$P(B \mid A)$ يُقرأ «احتمال B **بشرط** أن A قد وقع». معناه أننا نعرف مسبقًا أن A حدث، فنحصر أنفسنا داخل A فقط ونعتبره فضاء العينة الجديد.

الشرط دائمًا في المقام — $P(B \mid A) = P(A \cap B) \div P(A)$

مثال: احتمال أن يوقف سام سيارته في مكان ممنوع **و** يحصل على مخالفة 0.06 ، واحتمال أن يوقفها في مكان ممنوع 0.20 . إذا أوقفها فعلًا في مكان ممنوع، فاحتمال المخالفة = $0.06 \div 0.20 = 0.30$.

اختبار الاستقلال: إذا كان $P(B \mid A) = P(B)$ — أو $P(A \cap B) = P(A) \times P(B)$ — فالحدثان مستقلان، أي أن معرفة A لم تغيّر شيئًا.

كلمات الشرط في السؤال: given that، if، of those who، أو حقيقة مذكورة على أنها وقعت («وصل سام واضطر للوقوف في مكان ممنوع»).

EXAMPLE 4-32 · SELECTING COLORED CHIPS

A box contains black chips and white chips. A person selects two chips **without replacement**. If the probability of selecting a black chip and a white chip is 15/56, and the probability of selecting a black chip on the first draw is 3/8, find the probability of selecting the white chip on the second draw, *given* that the first chip selected was a black chip.

1. **Extract:** P(B ∩ W) = 15/56, P(B) = 3/8. Required: P(W | B).
2. $P(W \mid B) = \dfrac{P(B \cap W)}{P(B)} = \dfrac{15/56}{3/8} = \dfrac{15}{56} \times \dfrac{8}{3} = \dfrac{120}{168} = \dfrac{5}{7} \approx$ **0.714**

Dividing by a fraction: flip the bottom fraction and multiply. 15/56 ÷ 3/8 = 15/56 × 8/3. Cancel: 15/3 = 5 and 8/56 = 1/7 → 5/7.

EXAMPLE 4-33 · PARKING TICKETS

The probability that Sam parks in a no-parking zone *and* gets a parking ticket is 0.06, and the probability that Sam cannot find a legal parking space and has to park in the no-parking zone is 0.20. On Tuesday, Sam arrives at school and has to park in a no-parking zone. Find the probability that he will get a parking ticket.

1. **Extract:** Let N = parks in a no-parking zone, T = gets a ticket. P(N ∩ T) = 0.06, P(N) = 0.20.
2. "He *has to* park in a no-parking zone" = N has already happened → the question wants P(T | N).
3. $$P(T \mid N) = \frac{P(N \cap T)}{P(N)} = \frac{0.06}{0.20} = \boxed{0.30}$$

Why this is the exam's favourite shape: the condition is hidden inside a story ("he has to park there") rather than announced with the word "given". Learn to spot a condition stated as an accomplished fact.

5 · Two-way (contingency) tables — the complete toolkit

A two-way table lets you answer every question type in §4-1 to §4-3 by reading counts.
Master the four readings and this becomes free marks.

EXAMPLE 4-34 · SURVEY ON WOMEN IN THE MILITARY

A recent survey asked 100 people if they thought women in the armed forces should be permitted to participate in combat:

GENDER	YES	NO	TOTAL
Male	32	18	50
Female	8	42	50
Total	**40**	**60**	**100**

Find: (a) the respondent answered yes, given that the respondent was female; (b) the respondent was male, given that the respondent answered no; (c) are the events "Male" and "Yes" independent? (d) are "Male" and "Yes" mutually exclusive?

1. **(a) P(Yes | Female).** The condition "Female" restricts you to the Female row, total 50. Of those, 8 said yes.

   $$P(\text{Yes} \mid \text{Female}) = \frac{8}{50} = \boxed{0.16}$$

2. **(b) P(Male | No).** The condition "No" restricts you to the No column, total 60. Of those, 18 are male.

   $$P(\text{Male} \mid \text{No}) = \frac{18}{60} = \boxed{0.30}$$

3. **(c) Independent?** Test whether $P(M \cap Y) = P(M) \cdot P(Y)$.

   $$P(M \cap Y) = 32/100 = 0.32$$

   $$P(M) \cdot P(Y) = (50/100)(40/100) = 0.5 \times 0.4 = 0.20$$

   $0.32 \neq 0.20 \rightarrow$ **Not independent — the events are dependent.**

   Equivalent test: P(Y | M) = 32/50 = 0.64 but P(Y) = 40/100 = 0.40. Knowing the respondent is male changes the probability of "yes", so they are dependent.

4. **(d) Mutually exclusive?** $P(M \cap Y) = 32/100 = 0.32 \neq 0$. There are 32 people who are *both* male and said yes.

   → **Not mutually exclusive.**

> **PARTS (C) AND (D) TOGETHER ARE THE WHOLE POINT OF THE EXAMPLE**
>
> One table, two different questions that students merge into one. **Mutually exclusive** asks "*is the overlap empty?*" — look at the cell. **Independent** asks "*does one change the other?*" — compare a conditional with its unconditional version, or compare $P(A \cap B)$ with $P(A)P(B)$. Answering (d) with the independence test, or (c) with the overlap test, scores zero.

THE FOUR READINGS OF A TWO-WAY TABLE

QUESTION TYPE	WHAT GOES IN THE DENOMINATOR	EXAMPLE FROM 4-34
Simple P(A)	The grand total.	P(Male) = 50/100
Joint "and" P(A ∩ B)	The grand total; numerator is the single cell.	P(Male ∩ Yes) = 32/100
Union "or" P(A ∪ B)	The grand total; add the row and column, subtract the shared cell once.	P(Male ∪ Yes) = (50 + 40 − 32)/100 = 0.58
Conditional "given" P(A | B)	The *row or column total* of the condition.	P(Yes | Female) = 8/50

القاعدة الذهبية في جداول التقاطع

الشرط يحدد المقام. إذا قيل "given that it is a female" فالمقام هو مجموع صف الإناث (50) وليس المجموع الكلي (100). أما "and" فالمقام يبقى المجموع الكلي والبسط هو الخلية المشتركة.

و**فرّق دائمًا** بين: **mutually exclusive** (هل التقاطع فارغ؟ انظر إلى الخلية) و**independent** (هل يؤثر أحدهما في الآخر؟ قارن $P(A \cap B)$ مع $P(A) \cdot P(B)$).

6 · Probabilities for "at least" — the complement shortcut

This is the technique the lecture notes flag under "Other Examples: Probabilities for *At Least*" (textbook Examples 4-36 *Rolling a Die* and 4-37 *Ties*, set for self-study). It is the single most reusable trick in the chapter.

THE "AT LEAST ONE" RULE

$$P(\text{at least one}) = 1 - P(\text{none})$$

"At least one" has many ways to happen (exactly one, exactly two, exactly three, ...) but its complement — "none" — has exactly one. Compute the easy side and subtract.

الشرح بالعربي — «واحد على الأقل»

«واحد على الأقل» له طرق كثيرة للحدوث (واحد، اثنان، ثلاثة...)، لكن عكسه «ولا واحد» له طريقة **واحدة فقط**. لذلك:

$$P(\text{at least one}) = 1 - P(\text{none})$$

مثال: رمي نرد 3 مرات، احتمال ظهور 5 مرة واحدة على الأقل $= 1 - (5/6)^3 = 1 - 125/216 \approx 0.4213$.

كلما رأيت **at least** في السؤال، فكّر في المتمّمة قبل أي شيء آخر.

ILLUSTRATION · THE STANDARD SHAPE

A single die is rolled 3 times. Find the probability of getting **at least one** 5.

1. **The long way (don't):** P(exactly one 5) + P(exactly two 5s) + P(three 5s). Three separate calculations.
2. **The complement way:** "at least one 5" has complement "no 5s at all". P(not a 5 on one roll) = 5/6.
3. $P(\text{no 5s in 3 rolls}) = (5/6)^3 = \frac{125}{216}$
4. $P(\text{at least one 5}) = 1 - \frac{125}{216} = \frac{91}{216} \approx$ **0.4213**

The same pattern solves "at least one defective", "at least one female", "at least one match" — see Example 4-51(d) in §4-5, where it appears in its examined form.

THE THREE COMPLEMENT TRIGGERS

Reach for `1 − P(…)` the moment you read: **"at least one"** (complement = none) · **"not"** (complement = the event itself) · **"neither / none"** (complement = at least one). Scanning for these three phrases before you start writing will save you minutes on every paper.

4-4
4-5

Counting Rules, and Probability with Counting Rules

قواعد العدّ والاحتمال باستخدام قواعد العدّ

Everything so far needed `n(E)` and `n(S)`. When the sample space is too large to list, you *count* it instead. The four main counting rules are: the **Fundamental Counting Rule**, **Factorial Notation**, **Permutations**, and **Combinations**. These two sections carry In-Class Quiz II.

1 · The Fundamental Counting Rule

FUNDAMENTAL COUNTING RULE

In a sequence of n events in which the first has k_1 possibilities, the

$$k_1 \times k_2 \times \cdots \times k_n$$

الشرح بالعربي — قاعدة العدّ الأساسية

إذا كان لديك سلسلة من الاختيارات — الأول له k_1 طريقة، والثاني k_2، والثالث k_3 ... — فعدد كل الاحتمالات = **حاصل ضربها**.

مثال: 7 ألوان × 2 نوع × 3 ملمس × 2 استخدام = 84 نوع دهان.

- **مع التكرار** (**repetition allowed**): الأرقام تبقى كما هي من اليسار لليمين ← $10 \times 10 \times 10 \times 10 = 10{,}000$.
- **بدون تكرار** (**not allowed**): الأرقام تنقص واحدًا في كل خانة ← $10 \times 9 \times 8 \times 7 = 5{,}040$.

EXAMPLE 4-38 · TOSSING A COIN AND ROLLING A DIE

A coin is tossed and a die is rolled. Find the number of outcomes for the sequence of events.

$2 \times 6 =$ **12 outcomes**

Namely H1, H2, H3, H4, H5, H6, T1, T2, T3, T4, T5, T6.

EXAMPLE 4-39 · TYPES OF PAINT

A paint manufacturer wishes to manufacture several different paints. The categories include — Color: red, blue, white, black, green, brown, yellow. Type: latex, oil. Texture: flat, semigloss, high gloss. Use: outdoor, indoor. How many different kinds of paint can be made if you select one color, one type, one texture, and one use?

Count the options in each category: 7 colors, 2 types, 3 textures, 2 uses.

$7 \times 2 \times 3 \times 2 =$ **84 different kinds of paint**

EXAMPLE 4-40 · DISTRIBUTION OF BLOOD TYPES

There are four blood types: A, B, AB, O. Blood can also be Rh+ or Rh−. A blood donor can be classified as either male or female. How many different ways can a donor have his or her blood labeled?

$4 \times 2 \times 2 =$ **16 ways**

EXAMPLE 4-41 · RAILROAD MEMORIAL LICENSE PLATES

A license plate has a four-digit identification number. How many different plates can be made if: (a) digits are from 0 to 9 and repetitions are permitted? (b) digits are from 0 to 9 and repetitions are **not** permitted?

1. **(a) With repetition.** Every slot still has all 10 digits available:

   $10 \times 10 \times 10 \times 10 = 10^4 =$ **10,000 plates**

2. **(b) Without repetition.** Each digit used is removed from the pool:

   $10 \times 9 \times 8 \times 7 =$ **5,040 plates**

THE LECTURER'S OWN RULE OF THUMB

"If repetitions are permitted, then the numbers **stay the same** going from left to right. If repetitions are not permitted, then the numbers **decrease by 1** for each place left to right." Write this on your formula sheet exactly as stated — it answers half of all counting-rule questions on sight.

2 · Factorial notation

FACTORIAL

$$n! = n \times (n - 1) \times (n - 2) \times \cdots \times (2 \times 1)$$

$$0! = 1$$

Examples: $5! = 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 = 120$ · $9! = 9 \cdot 8 \cdot 7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 = 362{,}880$

الشرح بالعربي — المضروب

المضروب `n!` (يُقرأ **n factorial**) هو حاصل ضرب كل الأعداد من `n` نزولًا إلى `1`، مثل `5! = 5×4×3×2×1 = 120`.
وهو يساوي **عدد طرق ترتيب n أشياء مختلفة في صف**.

تعريف مهم جدًا: `0! = 1` وليس صفرًا.

FACTORIALS WORTH KNOWING COLD

N	0	1	2	3	4	5	6	7	8	9	10
n!	1	1	2	6	24	120	720	5,040	40,320	362,880	3,628,800

Casio keystrokes. To compute `(5 + 3)! = 40,320`: `( 5 + 3 )  SHIFT  x⁻¹ (x!)  =`

TRAP — 0! = 1, NOT 0

This matters because ${}_nP_n = n!/(n-n)! = n!/0! = n!$ and ${}_nC_n = n!/(0! \cdot n!) = 1$. If you set 0! = 0 you get a division by zero and lose the whole question. Memorise it as a definition, not as a calculation.

3 · Permutations — order matters

A **permutation** is an arrangement of *n* objects in a **specific order**.

PERMUTATION RULE 1 — N OBJECTS TAKEN R AT A TIME

$$ {}_nP_r = \frac{n!}{(n-r)!} $$

Casio: to compute ${}_7P_4$ → `7  SHIFT  ×  (nPr)  4  =`

الشرح بالعربي — التباديل (الترتيب مهم)

`nPr` = عدد طرق اختيار `r` من `n` **مع ترتيبها**. نستخدمها عندما يكون **الترتيب مهمًا**.

علامات أن الترتيب مهم: ترتيب أو تصنيف (**rank**)، مراكز (أول/ثاني/ثالث)، مناصب مختلفة (رئيس/نائب/أمين سر)، ترتيب الظهور، خانات مسمّاة (خريف/ربيع)، أرقام كود أو لوحة.

مثال: اختيار 3 ضيوف من 6 مع أهمية ترتيب ظهورهم = `6P3 = 6!/3! = 120`.

EXAMPLE 4-43 · BUSINESS LOCATION

(a) A business owner has a choice of 5 locations in which to establish her business. She decides to rank each location according to certain criteria, such as price of the store and parking facilities. How many different ways can she rank the 5 locations?
(b) Suppose she wishes to rank only the **top 3 locations** selected from 5 locations. How many different ways can she rank them?

1. **(a)** All 5 ranked, order matters: $5! = 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 =$ **120 ways**

   Equivalently: $_5P_5 = 5!/0! = 120/1 = 120$

2. **(b)** 3 chosen from 5, order matters: $_5P_3 = \dfrac{5!}{(5-3)!} = \dfrac{120}{2} =$ **60 ways**

EXAMPLE 4-44 · RADIO SHOW GUESTS

A radio talk show host can select 3 of 6 special guests for her program. The **order of appearance** of the guests is important. How many different ways can this be done?

"Order of appearance is important" → permutation.

$$_6P_3 = \frac{6!}{(6-3)!} = \frac{720}{6} =$$ **120 ways**

EXAMPLE 4-45 · SCHOOL MUSICAL PLAYS

A school musical director can select 2 musical plays to present next year. One will be presented in the fall, and one will be presented in the spring. If she has 9 to pick from, how many different possibilities are there?

The two slots are *distinguishable* (fall vs spring), so order matters → permutation.

$$_9P_2 = \frac{9!}{(9-2)!} = \frac{362{,}880}{5{,}040} =$$ **72 possibilities**

Why not a combination? Play X in fall + play Y in spring is a different programme from Y in fall + X in spring. Labelled slots always mean permutation.

EXAMPLE 4-42 · LOCK CODES

A person needs to make up a four-digit combination to open the lock on the garage door. The person decides to use four **different** digits from the first four digits, 0 through 3. How many different combinations can be made?

Four distinct digits arranged in all four positions — a full permutation:

$$4! = 4 \cdot 3 \cdot 2 \cdot 1 = \boxed{24 \text{ combinations}}$$

Note the vocabulary trap: everyday English calls this a "combination lock", but mathematically it is a *permutation* — 0123 and 3210 open different locks. The word "combination" in a question is never evidence; only the question "does order matter?" decides.

PERMUTATION RULE 2 — WHEN SOME OBJECTS ARE IDENTICAL

$$_nP_{r_1\ r_2\ r_3\ \dots\ r_p} = \frac{n!}{r_1! \cdot r_2! \cdot r_3! \cdots r_p!}$$

where $r_1 + r_2 + \cdots + r_p = n$. Divide out the arrangements you cannot tell apart.

الشرح بالعربي — تباديل مع عناصر متكررة

إذا كانت بعض الأشياء **متطابقة** (مثل الحروف المكررة في كلمة)، نقسم $n!$ على مضروب عدد تكرار كل عنصر، لأن تبديل عنصرين متطابقين لا يعطي ترتيبًا جديدًا.

مثال: كلمة STATISTICS فيها 10 حروف: S ثلاث مرات، T ثلاث مرات، I مرتان ← $10! \div (3! \times 3! \times 2!) = 50{,}400$.

نصيحة: عدّ الحروف أولًا وتأكد أن مجموع التكرارات يساوي n.

EXAMPLE 4-46 · LETTER PERMUTATIONS

How many permutations of the letters can be made from the word *STATISTICS*?

1. **Count the letters:** S-T-A-T-I-S-T-I-C-S → n = 10.
2. **Tally repeats:** S appears 3 times, T appears 3 times, I appears 2 times, A once, C once. Check: 3 + 3 + 2 + 1 + 1 = 10 ✔
3. $$\text{Permutations} = \frac{10!}{3! \cdot 3! \cdot 2! \cdot 1! \cdot 1!} = \frac{3{,}628{,}800}{6 \cdot 6 \cdot 2} = \frac{3{,}628{,}800}{72} =$$

HW 30 · TOOTHPASTE DISPLAY

How many different ways can 5 identical tubes of tartar control toothpaste, 3 identical tubes of bright white toothpaste, and 4 identical tubes of mint toothpaste be arranged in a grocery store counter display?

1. n = 5 + 3 + 4 = 12 tubes; three identical groups of sizes 5, 3, 4.
2. $$= \frac{12!}{5! \cdot 3! \cdot 4!} = \frac{479{,}001{,}600}{120 \cdot 6 \cdot 24} = \frac{479{,}001{,}600}{17{,}280} = \boxed{27{,}720 \text{ ways}}$$

4 · Combinations — order does not matter

A selection of distinct objects **without regard to order** is called a **combination**.

COMBINATION RULE

$$ {}_nC_r = \frac{n!}{(n-r)! \cdot r!} $$

Important: in a combination, the order or arrangement of the objects is *not* important; order *is* important in a permutation. Casio: to compute ${}_{10}C_4$ → `10 SHIFT ÷ (nCr) 4 =`

الشرح بالعربي — التوافيق (الترتيب غير مهم)

`nCr` = عدد طرق اختيار `r` من `n` **بدون ترتيب** — مجرد اختيار.

علامات أن الترتيب غير مهم: لجنة بدون مناصب، فريق، مجموعة، اختيار كتب أو أفلام أو مواضيع، سحب عدة قطع من صندوق دفعة واحدة.

العلاقة: `nCr = nPr ÷ r!` ، لذلك التوافيق دائمًا أصغر من أو تساوي التباديل.

مثال: اختيار 4 أفلام من 10 = `10C4 = 210` . وإذا كان الاختيار من مجموعتين (3 نساء من 7 **و** رجلان من 5) فإننا **نضرب**: `7C3 × 5C2 = 35 × 10 = 350` .

THE RELATIONSHIP WORTH SEEING ONCE

${}_nC_r = {}_nP_r \div r!$ — a combination is a permutation with the $r!$ internal orderings divided out. That is why ${}_nC_r$ is always ≤ ${}_nP_r$. If your combination answer comes out *larger* than the matching permutation, you have used the wrong formula.

EXAMPLE 4-48 · MOVIES AT THE PARK

The director of Movies at the Park must select 4 movies from a total of 10 movies to show on Movie Night at the Park. How many different ways can the selections be made?

Just a selection — nothing says the four are shown in a particular order → combination.

$$ {}_{10}C_4 = \frac{10!}{(10-4)! \cdot 4!} = \frac{10 \cdot 9 \cdot 8 \cdot 7}{4 \cdot 3 \cdot 2 \cdot 1} = \frac{5{,}040}{24} = \boxed{210 \text{ ways}} $$

EXAMPLE 4-49 · COMMITTEE SELECTION

In a club there are 7 women and 5 men. A committee of 3 women and 2 men is to be chosen. How many different possibilities are there?

1. **Choose the women:** ${}_7C_3$ = 7!/(4!·3!) = (7·6·5)/(3·2·1) = 210/6 = **35**
2. **Choose the men:** ${}_5C_2$ = 5!/(3!·2!) = (5·4)/2 = **10**
3. **Combine with the Fundamental Counting Rule** (a choice of women *and* a choice of men):

   $35 \times 10 =$ **350 possibilities**

TRAP — "AND" MEANS MULTIPLY, NOT ADD

Choosing 3 women **and** 2 men = $35 \times 10 = 350$. Students who write 35 + 10 = 45 are answering a different question ("choose 3 women *or* 2 men"). Within counting rules, **and → ×**, **or → +**.

5 · Summary of the counting rules (Table 4-1)

TABLE 4-1 — SUMMARY OF COUNTING RULES

RULE	DEFINITION	FORMULA
Fundamental counting rule	The number of ways a sequence of n events can occur if the first event can occur in k_1 ways, the second in k_2 ways, etc.	$k_1 \cdot k_2 \cdot k_3 \cdots k_n$
Permutation rule 1	The number of permutations of n objects taking r objects at a time (**order is important**).	${}_nP_r = n! / (n-r)!$

RULE	DEFINITION	FORMULA
Permutation rule 2	The number of permutations of n objects when r_1 objects are identical, r_2 objects are identical, ..., r_p objects are identical.	$n! / (r_1!\ r_2!\ \cdots\ r_p!)$
Combination rule	The number of combinations of r objects taken from n objects (**order is not important**).	$_nC_r = n! / [(n - r)!\ r!]$

6 · §4-5 — Probability and Counting Rules

This is the destination. The counting rules supply `n(E)` and `n(S)`; classical probability divides them. The structure of every §4-5 question is identical:

STEP 1 **Count the sample space.**
How many total ways can the selection be made? Usually a single $_nC_r$ or a product.

STEP 2 **Count the event.**
How many of those ways satisfy the condition? Usually a product of smaller `C`'s — one per category.

STEP 3 **Divide.**
`P(E) = n(E) / n(S)`, then round sensibly and interpret in words.

الشرح بالعربي — الاحتمال باستخدام قواعد العدّ (أهم نوع أسئلة)

كل سؤال من هذا النوع له نفس الخطوات الثلاث:

1. **احسب `n(S)`:** عدد كل طرق الاختيار الممكنة — غالبًا `nCr` واحدة.
2. **احسب `n(E)`:** عدد الطرق التي تحقق الشرط — غالبًا حاصل ضرب توافيق، واحدة لكل فئة.
3. **اقسم:** `P(E) = n(E) ÷ n(S)`، ثم اكتب جملة تفسيرية.

مثال: 24 قطعة منها 4 معيبة، نختار 4. احتمال قطعتين معيبتين بالضبط = `(4C2 × 20C2) ÷ 24C4 = 1140 ÷ 10626 ≈ 0.1073`. وإذا طُلب «واحدة معيبة على الأقل» ← `1 − P(none) = 1 − 0.4560 = 0.5440`.

قاعدة الاتساق: استخدم نفس نوع العدّ في البسط والمقام — لا تضع تباديل فوق توافيق.

EXAMPLE 4-51 · DEFECTIVE INTEGRATED CIRCUITS

A box contains 24 integrated circuits, 4 of which are defective. If 4 are sold at random, find: (a) exactly 2 are defective; (b) none is defective; (c) all are defective; (d) at least 1 is defective.

Set-up. 4 defective, 20 good, 24 total. Choosing 4 with no order → combinations.

$$n(S) = {}_{24}C_4 = \frac{24 \cdot 23 \cdot 22 \cdot 21}{4 \cdot 3 \cdot 2 \cdot 1} = \frac{255{,}024}{24} = \mathbf{10{,}626}$$

1. **(a) Exactly 2 defective** — so 2 from the 4 defective *and* 2 from the 20 good:

   $$n(E) = {}_4C_2 \times {}_{20}C_2 = 6 \times 190 = 1{,}140$$

   $$P = \frac{1{,}140}{10{,}626} \approx \mathbf{0.1073}$$

2. **(b) None defective** — all 4 from the 20 good:

   $$n(E) = {}_{20}C_4 = \frac{20 \cdot 19 \cdot 18 \cdot 17}{24} = 4{,}845$$

   $$P = \frac{4{,}845}{10{,}626} \approx \mathbf{0.4560}$$

3. **(c) All defective** — all 4 from the 4 defective:

   $$n(E) = {}_4C_4 = 1$$

   $$P = \frac{1}{10{,}626} \approx \mathbf{0.000094}$$

4. **(d) At least 1 defective** — the complement of "none defective", which is part (b):

   $$P = 1 - 0.4560 = \mathbf{0.5440}$$

PART (D) IS THE EXAM'S REWARD FOR RECOGNISING THE COMPLEMENT

Done the long way, (d) = P(1 defective) + P(2) + P(3) + P(4) — four calculations. Done as `1 − P(none)`, it is one subtraction you have already computed. This is precisely why parts (b) and (d) appear in the same question: the examiner is testing whether you noticed.

EXAMPLE 4-52 · TERM PAPER SELECTION

A student needs to select two topics to write two term papers for a course. There are 8 topics in economics and 11 topics in science. Find the probability that she selects one topic in economics and one topic in science.

1. **n(S):** any 2 of the 19 available topics → $_{19}C_2 = \frac{19 \cdot 18}{2} = 171$
2. **n(E):** one economics *and* one science → $_8C_1 \times {}_{11}C_1 = 8 \times 11 = 88$
3. $P = \frac{88}{171} \approx$ **0.5146**

EXAMPLE 4-53 · STATE LOTTERY NUMBER

In the Pennsylvania State Lottery, a person selects a three-digit number and repetitions are permitted. If a winning number is selected, find the probability that it will have all three digits the same.

1. **n(S):** three digits, repetition permitted → $10 \times 10 \times 10 = 1{,}000$
2. **n(E):** all three digits the same → 000, 111, 222, ..., 999 → 10
3. $P = \frac{10}{1{,}000} =$ **0.01** (1%)

EXAMPLE 4-54 · TENNIS TOURNAMENT

There are 8 married couples in a tennis club. If 1 man and 1 woman are selected at random to plan the summer tournament, find the probability that they are married to each other.

1. **n(S):** 8 men to choose from *and* 8 women → $8 \times 8 = 64$ pairs
2. **n(E):** only 8 of those pairs are actually married to each other (one per couple) → 8
3. $P = \frac{8}{64} = \frac{1}{8} =$ **0.125**

Alternative reasoning that proves the answer: pick any man first; his wife is 1 of the 8 women, so P = 1/8 directly. ✔

HW 11 · SOCKS IN A DRAWER

A drawer contains 11 identical red socks and 8 identical black socks. Suppose you choose 2 socks at random in the dark. (a) What is the probability that you get a pair of red socks? (b) A pair of black socks? (c) 2 unmatched socks?

n(S): any 2 of 19 socks → $_{19}C_2 = \frac{19 \cdot 18}{2} = 171$

1. **(a) Pair of red:** $_{11}C_2 = \frac{11 \cdot 10}{2} = 55 \rightarrow P = \frac{55}{171} \approx$ **0.3216**
2. **(b) Pair of black:** $_{8}C_2 = \frac{8 \cdot 7}{2} = 28 \rightarrow P = \frac{28}{171} \approx$ **0.1637**
3. **(c) 2 unmatched:** one red *and* one black: $_{11}C_1 \times {}_{8}C_1 = 88 \rightarrow P = \frac{88}{171} \approx$ **0.5146**

   Or by complement: 1 − 0.3216 − 0.1637 = 0.5146 ✔

Check: 55 + 28 + 88 = 171 = n(S). The three cases are exhaustive and mutually exclusive, so their counts must total the sample space. ✔

القاعدة الحاسمة: Permutation أم Combination؟

اسأل: **هل يتغير الناتج إذا غيّرت الترتيب؟**

- **نعم** (ترتيب، تصنيف، مراكز أول/ثاني، مناصب مختلفة، أرقام لوحة) ← **Permutation** `nPr`
- **لا** (لجنة، مجموعة، اختيار عدد من أشياء بدون أدوار) ← **Combination** `nCr`

وتذكّر: كلمة "**combination lock**" في اللغة الدارجة هي في الحقيقة **permutation** رياضيًا، فلا تعتمد على الكلمة — اعتمد على السؤال.

FS

Master formula sheet

ورقة القوانين الشاملة — احفظها كاملة

Everything in Chapter 4 on one page. If you can reproduce this from memory in four minutes, you are ready.

كيف تستخدم ورقة القوانين

اقرأها حسب القسم (الألوان نفسها في الملخص)، والعمود الأخير يخبرك **متى** تستخدم كل قانون، وتحت كل اسم إنجليزي اسمه بالعربي. التدريب الحقيقي: اكتبها من ذاكرتك على ورقة فارغة، ثم قارن وصحّح بالأحمر. إذا استطعت كتابتها كاملة في أربع دقائق فأنت جاهز.

NAME	FORMULA	WHEN
§4-1 · Sample spaces and probability		
Classical probability الاحتمال الكلاسيكي	`P(E) = n(E) / n(S)`	Equally like outcomes; you can cou the sample space.
Empirical probability الاحتمال التجريبي	`P(E) = f / n`	A frequency distribution or survey data is give
Probability range مدى الاحتمال	`0 ≤ P(E) ≤ 1`	Always. Use to sanity-check every answer.
Total probability مجموع الاحتمالات	`Σ P(all outcomes in S) = 1`	Check on ar frequency c tree table.
Complement rule قاعدة المتمّمة	`P(Ē) = 1 − P(E)`	"not", "neither", "none", "at least one".
§4-2 · Addition rules — "OR"		

NAME	FORMULA	WHEN
Mutually exclusive test اختبار التنافي	$A \cap B = \varnothing$	Can the two events happen together? N → mutually exclusive.
Addition rule 1 قاعدة الجمع الأولى	$P(A \cup B) = P(A) + P(B)$	A and B **are** mutually exclusive.
Addition rule 2 قاعدة الجمع الثانية	$P(A \cup B) = P(A) + P(B) - P(A \cap B)$	A and B are **not** mutuall exclusive. Safe in both cases.
Three mutually exclusive ثلاثة أحداث متنافية	$P(A \cup B \cup C) = P(A) + P(B) + P(C)$	Three disjoi events.
Three general ثلاثة أحداث غير متنافية	$P(A \cup B \cup C) = \Sigma P(\text{single}) - \Sigma P(\text{pairs}) + P(A \cap B \cap C)$	Three overlapping events.
§4-3 · Multiplication rules — "AND" and "GIVEN"		
Multiplication rule 1 قاعدة الضرب الأولى	$P(A \cap B) = P(A) \cdot P(B)$	A and B independen (with replacemer separate devices).

NAME	FORMULA	WHEN
Multiplication rule 2 قاعدة الضرب الثانية	$P(A \cap B) = P(A\|B)\cdot P(B) = P(B\|A)\cdot P(A)$	A and B dependent (without replacemer "of those who...").
Conditional probability الاحتمال الشرطي	$P(B \mid A) = P(A \cap B) / P(A)$	"given that" "if", "of thos who", a stated accomplish fact.
Independence test اختبار الاستقلال	$P(B\|A) = P(B)$ or $P(A \cap B) = P(A)\cdot P(B)$	Asked directly, especially from a two-way table.
At least one واحد على الأقل	$P(\text{at least one}) = 1 - P(\text{none})$	Any questic containing least".
§4-4 & §4-5 · Counting rules		
Fundamental counting rule قاعدة العدّ الأساسية	$k_1 \times k_2 \times \cdots \times k_n$	A sequence of independer choices; repetition allowed or not.
Factorial المضروب	$n! = n(n-1)(n-2)\cdots(2)(1), \quad 0! = 1$	Arranging a n objects.
Permutation rule 1 التباديل	${}_nP_r = n! / (n - r)!$	Choose r from n, **ord matters**.

NAME	FORMULA	WHEN
Permutation rule 2 التباديل مع عناصر متكررة	$n! / (r_1!\ r_2!\ \cdots\ r_p!)$	Arrange n objects, some identical (letters, identical items).
Combination rule التوافيق	${}_nC_r = n! / [(n - r)!\ r!]$	Choose r from n, **ord does not matter**.
Link between them العلاقة بينهما	${}_nC_r = {}_nP_r / r!$	Sanity chec C is always P.
Probability by counting الاحتمال بالعدّ	P(E) = n(E) / n(S), both counted with the rules above	**The signature 5 question.**

Casio calculator keystrokes

TO COMPUTE	PRESS	RESULT
(5 + 3)!	(5 + 3) SHIFT x^{-1} (x!) =	40,320
${}_7P_4$	7 SHIFT × (nPr) 4 =	840
${}_{10}C_4$	10 SHIFT ÷ (nCr) 4 =	210

DT Which rule do I use?

كيف أختار القانون الصحيح خلال عشر ثوانٍ

Run this in order on every question. It takes ten seconds and it is the difference between a confident answer and a guess.

Q1 **Does the question contain "not", "neither", "none", or "at least"?**

→ Yes: use the complement, `1 − P(opposite)`. Compute the easy side. This check comes first because it can collapse the whole problem.

Q2 **Does it say "given that", "if", "of those who", or state a fact as already settled?**

→ Yes: conditional probability, $P(B|A) = P(A \cap B)/P(A)$. From a table, the condition's row or column total is the denominator.

Q3 **Does it say "or"?**

→ Addition rule. Then ask: can both happen together? No → $P(A)+P(B)$. Yes → $P(A)+P(B)-P(A \cap B)$.

Q4 **Does it say "and", or describe a sequence of stages?**

→ Multiplication rule. Then ask: with replacement / separate devices? → independent, $P(A) \cdot P(B)$. Without replacement / "of those"? → dependent, $P(A) \cdot P(B|A)$.

Q5 **Is the sample space too big to list?**

→ Counting rules. Then ask: does order matter? Yes → ${}_nP_r$. No → ${}_nC_r$. Several categories combined → multiply them (Fundamental Counting Rule).

Q6 **Is it asking for a probability after all that counting?**

→ $P(E) = n(E)/n(S)$. Count the event and the sample space with the same kind of rule — never mix a permutation numerator with a combination denominator.

خطوات اختيار القانون بالعربي — طبّقها بالترتيب

1. هل في السؤال **at least** / **none** / **neither** / **not**؟ ← استخدم المتمّمة `P(العكس) = 1 −` .
2. هل فيه **given that** / **if** / **of those who**، أو حقيقة وقعت مسبقًا؟ ← احتمال شرطي، والشرط في المقام.
3. هل فيه **or**؟ ← قاعدة الجمع. هل يمكن وقوعهما معًا؟ لا ← اجمع. نعم ← اجمع واطرح التقاطع.
4. هل فيه **and** أو مراحل متتالية؟ ← قاعدة الضرب. مع إرجاع ← اضرب مباشرة. بدون إرجاع ← الاحتمال الثاني يتغيّر.
5. هل فضاء العينة أكبر من أن يُسرد؟ ← قواعد العدّ. الترتيب مهم ← `nPr` . غير مهم ← `nCr` . عدة فئات ← اضرب.
6. هل المطلوب احتمال بعد العدّ؟ ← `n(E) ÷ n(S)` ، بنفس نوع العدّ في البسط والمقام.

THE ONE CONSISTENCY RULE THAT CATCHES MOST §4-5 ERRORS

Whatever you use for `n(S)` , use for `n(E)` . If the sample space counts *unordered* selections (combinations), the event must count unordered selections too. A numerator built from permutations over a denominator built from combinations is the most common wrong answer in the whole chapter, and it is always too large by a factor of `r!` .

TR The 14 traps that cost marks

الأخطاء الأربعة عشر التي تضيّع الدرجات

Each of these has a specific home in this chapter. Read the list the night before the exam.

#	THE TRAP	WHAT TO DO INSTEAD
1	Confusing **mutually exclusive** with **independent**.	Mutually exclusive = no overlap (`A∩B=∅`). Independent = no influence (`P(A∩B)=P(A)P(B)`). Two different questions, two different tests.
2	Using Addition Rule 1 when the events overlap.	Always test the intersection first. When in doubt use Rule 2; it reduces to Rule 1 automatically.
3	Reading "or" as exclusive.	In probability, "A or B" always means A, or B, *or both*.
4	Missing "**without replacement**".	Circle it. It changes every denominator after the first draw and switches you from Rule 1 to Rule 2.

#	THE TRAP	WHAT TO DO INSTEAD
5	Putting the wrong thing in the conditional denominator.	The *condition* is the denominator. `P(B|A)` divides by `P(A)`, never by `P(B)`.
6	Treating "of these clients..." / "of those who..." as unconditional.	That phrasing always hands you a conditional probability. Annotate it as `P(A|H)` before you calculate.
7	Mixing up "at most" and "less than" (and "at least" / "more than").	**at** → the boundary value is *included*. **than** → the boundary value is *excluded*.
8	Doing "at least one" the long way.	`1 − P(none)`. Always.
9	Using a permutation for a committee.	A committee has no internal order → combination. Labelled roles (chair, secretary) → permutation.
10	Adding instead of multiplying across categories.	"3 women **and** 2 men" → ${}_7C_3 \times {}_5C_2$. "And" is multiplication.
11	Writing 0! = 0.	`0! = 1`, by definition. It appears in ${}_nP_n$ and ${}_nC_n$.
12	Mixing permutations and combinations between `n(E)` and `n(S)`.	Count both with the same rule. Consistency, not cleverness.
13	Giving only the final number.	Write the sample space in set notation, name the rule you are using, show the substitution. The lecturer states explicitly that set notation is required.
14	Reporting a probability greater than 1 or negative.	$0 \le P \le 1$. If your answer falls outside, you have added when you should have multiplied, or forgotten to divide by `n(S)`. Catch it before you move on.

الأخطاء الأربعة عشر بالعربي

1. الخلط بين **متنافيين** (لا تداخل) و**مستقلين** (لا تأثير).
2. الجمع المباشر مع أحداث متداخلة — اطرح التقاطع.
3. فهم **or** على أنها «أحدهما فقط» — هي «أحدهما أو كلاهما».
4. تجاهل **without replacement** — المقام يتغير في السحبة الثانية.
5. وضع الشيء الخطأ في مقام الاحتمال الشرطي — **الشرط هو المقام**.
6. اعتبار **of those who** احتمالًا عاديًا — هو احتمال شرطي.
7. الخلط بين **at most** و **less than** — كلمة **at** تعني أن الرقم **داخل** الحساب.
8. حساب **at least one** بالطريقة الطويلة — استخدم `1 - P(none)`.
9. استخدام التباديل للجان — اللجنة بدون مناصب = **توافيق**.
10. الجمع بدل الضرب عند الاختيار من فئتين — **and** تعني **ضرب**.
11. كتابة `0! = 0` — الصحيح `0! = 1`.
12. خلط التباديل والتوافيق بين البسط والمقام.
13. كتابة الناتج النهائي فقط — اكتب فضاء العينة واسم القانون والتعويض.
14. احتمال أكبر من 1 أو سالب — يعني أن في الحل خطأ؛ راجعه قبل أن تنتقل.

RV The review self-study exercises, fully solved

تمارين المراجعة محلولة بالكامل

These are the three exercises Dr. Al-Hadhrami closes the chapter with, set for self-study. They are solved here step by step.

الشرح بالعربي

هذه تمارين المراجعة الثلاثة في آخر ملف الدكتور. الفكرة المشتركة بينها: **ارسم مخطط فن واملأ المناطق الأربع بالأرقام** (A فقط، B فقط، المشترك، خارج الاثنين)، ثم يصبح كل فرع مجرد جمع لمناطق تراها أمامك. وتذكّر أن **neither ... nor** تعني دائمًا `1 - P(A ∪ B)`.

EX 1 · BURGERS AND PIZZA

A study shows that about 37 percent of school children eat burger, 17 percent eat pizza, and 10 percent eat both burger and pizza. What is the probability that a randomly selected school child eats **neither** burger nor pizza?

1. **Extract:** P(B) = 0.37, P(Z) = 0.17, P(B ∩ Z) = 0.10.
2. **"Neither…nor" is the complement of the union.** First get the union — they overlap, so Rule 2:

   $P(B \cup Z) = 0.37 + 0.17 - 0.10 = 0.44$

3. $P(\text{neither}) = 1 - P(B \cup Z) = 1 - 0.44 =$ **0.56**

Cross-check with a Venn diagram: burger only = 0.37 − 0.10 = 0.27; pizza only = 0.17 − 0.10 = 0.07; both = 0.10; outside = 1 − (0.27 + 0.07 + 0.10) = 0.56. ✔

EX 2 · SMART PHONE COLORS

If the probabilities are, respectively, 0.23, 0.15, 0.21, and 0.09 that a person purchasing a new smart phone will choose the color gold, silver, black, or white, what is the probability that a given buyer will purchase a new phone that comes in one of those colors?

1. A phone has exactly one color, so the four events are **mutually exclusive** → extended Addition Rule 1.
2. $P(\text{gold} \cup \text{silver} \cup \text{black} \cup \text{white}) = 0.23 + 0.15 + 0.21 + 0.09 =$ **0.68**

Read what this tells you: the total is 0.68, not 1 — so 32% of buyers choose some other color entirely. A question that follows up with "find the probability the buyer chooses a different color" is answered by 1 − 0.68 = 0.32.

EX 3 · SET ALGEBRA WITH PROBABILITIES

If $P(A) = 0.3$, $P(B) = 0.2$, and $P(A \cap B) = 0.1$, determine: (a) $P(A^c)$ (b) $P(A \cup B)$ (c) $P(A^c \cap B)$ (d) $P(A \cap B^c)$ (e) $P((A \cup B)^c)$ (f) $P(A^c \cup B)$

Build the four-region picture first. A Venn diagram with two overlapping circles has exactly four regions, and every part of this question reads straight off it:

REGION	MEANING	PROBABILITY
A only	$A \cap B^c$	$0.3 - 0.1 = 0.2$
Both	$A \cap B$	0.1
B only	$A^c \cap B$	$0.2 - 0.1 = 0.1$
Neither	$(A \cup B)^c$	$1 - 0.4 = 0.6$
Total	S	**1.0** ✔

1. **(a)** $P(A^c) = 1 - P(A) = 1 - 0.3 =$ **0.7**
2. **(b)** $P(A \cup B) = P(A) + P(B) - P(A \cap B) = 0.3 + 0.2 - 0.1 =$ **0.4**
3. **(c)** $P(A^c \cap B)$ = "B but not A" $= P(B) - P(A \cap B) = 0.2 - 0.1 =$ **0.1**
4. **(d)** $P(A \cap B^c)$ = "A but not B" $= P(A) - P(A \cap B) = 0.3 - 0.1 =$ **0.2**
5. **(e)** $P((A \cup B)^c) = 1 - P(A \cup B) = 1 - 0.4 =$ **0.6**
6. **(f)** $P(A^c \cup B) = P(A^c) + P(B) - P(A^c \cap B) = 0.7 + 0.2 - 0.1 =$ **0.8**
   Or from the diagram: everything except "A only" $= 1 - 0.2 = 0.8$ ✔

THE TECHNIQUE THAT MAKES THIS QUESTION TRIVIAL

Never attack set-algebra questions with formulas alone. Draw the two circles, fill in the four regions as numbers, and every sub-part becomes an addition of regions you can see. Parts (c), (d) and (f) take longer to write than to solve once the diagram exists.

A+

The A+ study protocol

بروتوكول الطالب الأول — استراتيجيات الوصول إلى A+

Chapter 4 rewards a specific kind of preparation: not more hours, but the right sequence. This is what the top student in a STAT 2101 section actually does, in order.

Phase 1 — Build the skeleton (90 minutes, once)

1.1 **Write the formula sheet by hand, from memory, with the page closed.**
Then open it and mark every gap in red. Repeat two days later. Recall is what the exam measures; re-reading is not recall and feels far more productive than it is.

1.2 **Make a one-page decision tree in your own words.**
Copy the "Which rule do I use?" section above, but phrase every branch yourself. The act of rewording is what moves it into memory.

1.3 **Learn the definitions as sentences, not as feelings.**
Sample space, outcome, event, simple, compound, mutually exclusive, independent, dependent, complement. These are CLO1 marks; they appear on quizzes verbatim and take one minute each to secure.

Phase 2 — Drill by question type, not by page order

2.1 **Group every exercise by which rule solves it**
before solving any of them. Fifteen minutes of sorting teaches you more about rule selection than two hours of grinding, because rule selection — not arithmetic — is where marks are lost.

2.2 **Do the assigned exercises properly: p. 247, numbers 6, 8, 16 and 40.**
These are named in the course outline, which means the tutor has them, which means variants of them are the most likely quiz questions in the course. Then do the Moodle exercises — they are listed for both Week 5 and Week 6.

2.3 **Solve every "Self-study" example in the slide deck.**
Examples 4-13, 4-21, 4-22, 4-26, 4-28, 4-40, 4-45, 4-54 and the three review exercises are marked self-study, which means they are *not* done in class and are therefore exactly where an examiner finds unfamiliar-looking material. All of them are solved in this document — close it and redo them from the question alone.

2.4 **Build the "twin" of each problem you solve.**
Did it say *with* replacement? Redo it *without*. Did it ask for "exactly 2"? Also do "at least 1". Twin-building is the single highest-yield habit in this chapter because the examiner builds twins for exactly the same reason.

Phase 3 — Write like a marker reads

3.1 **Open every answer by defining your events in symbols.**
"Let N = the staff person is a nurse, M = the staff person is male." Two lines, and the rest of the solution writes itself — and the marker can follow you even if the arithmetic slips.

3.2 **Write the sample space in set notation whenever it is small enough.**
The lecturer flags this explicitly on the Example 4-6 slide: "Sample spaces and events need to be expressed using set notations." That is a marking instruction in disguise.

3.3 **Name the rule before you use it.**
"Since the events are not mutually exclusive, Addition Rule 2 applies." One sentence, often one mark, and it forces you to check your own choice.

3.4 **Finish with a sentence in words.**
"So there is about a 54% chance that at least one circuit is defective." Interpretation is a stated outcome of this course (CLO8) and costs five seconds.

Phase 4 — The exam hour itself

4.1 **Brain-dump the formula sheet onto the back page in the first 90 seconds**
, before reading any question. You are then calculating from a written reference instead of from a memory under pressure.

4.2 **Underline the decisive words as you read:**

and, or, not, at least, at most, given, without replacement, order is important. Every one of them selects a formula. The question is telling you which rule to use; most lost marks come from not listening.

4.3 **Run the three checks on every numerical answer:**

is it between 0 and 1? do the complementary parts sum to 1? does the magnitude feel right (a rare event should come out small)? These catch nearly every slip.

4.4 **Budget by marks, not by difficulty.**

The midterm is 60 minutes across Chapters 1–4. Chapter 4 questions look longest but have the most structured method — do them with a timer at home so you know your real pace before exam night.

استراتيجية الطالب الأول — باختصار

- **اكتب ورقة القوانين من ذاكرتك** ثم صحّحها. الاسترجاع هو ما يُختبر، وليس القراءة.
- **صنّف التمارين حسب القانون المستخدم** قبل حلّها — ضياع الدرجات يكون في اختيار القانون لا في الحساب.
- **حلّ كل الأمثلة المكتوب عليها Self-study** فهي غير محلولة في المحاضرة، ومنها تأتي الأسئلة غير المألوفة.
- **اصنع "التوأم" لكل مسألة:** إذا كانت with replacement فأعدها without، وإذا كانت "exactly 2" فأعدها "at least 1".
- **في الامتحان:** اكتب القوانين في أول 90 ثانية، ضع خطًا تحت الكلمات المفتاحية (/ and / or / not / at least / given without replacement)، وتأكد أن كل إجابة بين 0 و 1.
- **اكتب جملة تفسيرية في النهاية** — "أي أن احتمال وجود قطعة معيبة واحدة على الأقل يقارب 54%". هذه درجات شبه مجانية.

The fastest wins, ranked

RANK	ACTION	TIME COST
1	Memorise the five phrase translations (exactly / less than / at most / more than / at least).	10 minutes
2	Memorise `P(at least one) = 1 − P(none)` and scan every question for "at least".	5 minutes

RANK	ACTION	TIME COST
3	Drill the permutation-vs-combination decision on 20 one-line prompts.	25 minutes
4	Practise building a two-way table from a word problem (Example 4-21 is the template).	20 minutes
5	Learn the definitions verbatim for CLO1 marks.	20 minutes
6	Rehearse the Casio keystrokes for `x!`, `nPr`, `nCr` until they are automatic.	10 minutes

That is ninety minutes of work that covers the majority of the marks available in Chapter 4. Everything else is practice volume.

STAT 2101 — Introduction to Statistics · Chapter 4: Probability and Counting Rules. Rebuilt from Dr. Ahmed Al-Hadhrami's lecture slides (Spring 24) and aligned to the SQU College of Science course outline, OQF Level 5, Fall 2026. Textbook: Allan G. Bluman, *Elementary Statistics: A Step by Step Approach*, brief version, 10th edition, McGraw Hill (2017).

Every example, definition, formula and note from the source deck is reproduced here, with full worked solutions supplied for the examples the slides leave open. A companion question bank with full solutions accompanies this document.